# Brain Tumor Detection and Classification Using PSO and SVM Classifier

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**Abstract: The purpose of this project it to identify the brain tumor at early stage which is very difficult task for doctors to identify. MRI images are more prone to noise and other environmental interference. So, it becomes difficult for doctors to identify tumor and their causes. So here we come up with the system, where system will detect brain tumor from images. The scope is to develop an automated system for enhancement, segmentation and classification of brain tumors. The system can be used by neurosurgeons and healthcare specialists. The system incorporates image processing, image segmentation, feature extraction and image classification and is expected to improve the sensitivity, specificity, and efficiency of brain tumor detection.**

**Keywords— Brain tumor; MRI images; PSO; SVM; classification;**

## I.INTRODUCTION

Image Processing has an important role in medical field and medical imaging is a challenging field. Medical imaging has advantage in diagnosis of the disease. Medical imaging provides proper diagnosis of brain tumor. There are many techniques to detect brain tumor from MRI images.

Brain tumor is an abnormal and undesirable growth of tissue cells in brain which leads to certain neurological disorder for human. Now a days due to environmental and human life style these tumor cases are rapidly increasing. There are two main types of tumors: malignant (cancerous) tumors and benign (non-cancerous) tumors. Cancerous tumors can be divided into primary tumors, which start within the brain, and secondary tumors, which have spread from elsewhere, known as brain metastasis tumors. There are over 120 different types of primary brain tumors. A benign brain tumor consists of very slow-growing cells, usually has distinct borders and rarely spreads. When viewed under a microscope, these cells have an almost normal cellular appearance. A malignant brain tumor is usually rapidly-growing, invasive and life-threatening. Malignant brain tumors are sometimes called brain cancer. The purpose of the project being identifying brain tumor at early stage which is very difficult task for doctors to identify. MRI images are more prone to noise and other environmental interference. So, it becomes difficult for doctors to identify tumor and their causes. So here we come up

with the system, where system will detect brain tumor from images. Here we convert image into grayscale image. User has to select the image. System will process the image by applying image processing steps. We applied a unique algorithm to detect tumor from brain image.

## II. RELATED WORK

In [1] they proposed Firstly image is acquired then it Is subjected to pre-processing then k-means clustering algorithm is used to separate the images into 4 clusters of varying intensity levels, then the cluster with the tumor region is found then the tumor is extracted using morphological & region properties operation. The proposed method is divided into 3 sections: image acquisition, pre-processing and post-processing. The k-means clustering algorithm is used on the filtered image with k=4 clusters. This divides the image into four levels of grayscale intensity and separated the tumor and skull from the brain matter, background and cerebrospinal fluid. It was found that the algorithm worked well for 14 out of the 23 images in the database. This indicates a robustness of 60% in extracting only the tumor from the image.

In [2] system works on medical image dataset using data mining technique. This system identifies the type of tumor by support vector machine, here we use genetic optimization algorithm for optimizing the features. System classification performance will be improved by feature weighting and SVM parameter optimization. Here we simultaneously optimize the SVM parameter and feature by genetic algorithm. Pre-processing and skull removal process will increase the performance of the system. This System will provide better accuracy with GA-SVM Classifier and will increase the decision making capacity.

In [3] the process which is followed in classification of brain tumor is very complicated. There are various steps for the medical image processing like image segmentation, image extraction and image classification. Different types of features are extracted from the segmented MRI images like intensity, shapes and texture based features. The feature selection approach is used to select the small subset of features from MRI image which minimize redundancy and maximize relevance to the target. In this paper, online database of MRI images containing brain tumor is taken then a machine learning model is developed by using the Particle Swarm Optimization (PSO) algorithm for feature selection and then Support Vector Machine (SVM) classifier is used to classify the type of tumor in present brain MRI images.

In [6] they proposed K-means is the backbone of the paper methodology. k-means algorithm starts clustering by determining k initial points either at random or using some heuristic data, it then groups each image pixel under the central point it is closest to. Next it calculates new central points and repeat the former two steps. Clustering refers to the process of grouping pixels of an image such that pixels which are in the same group (cluster) are similar among them and are dissimilar to the pixels which belong to the other groups (clusters). the number of tumor pixels, K-means clustering gave a better result than the other methods. The clustering algorithm was tested with a database of 100 MRI brain images. K-means clustering achieved about 95% result.

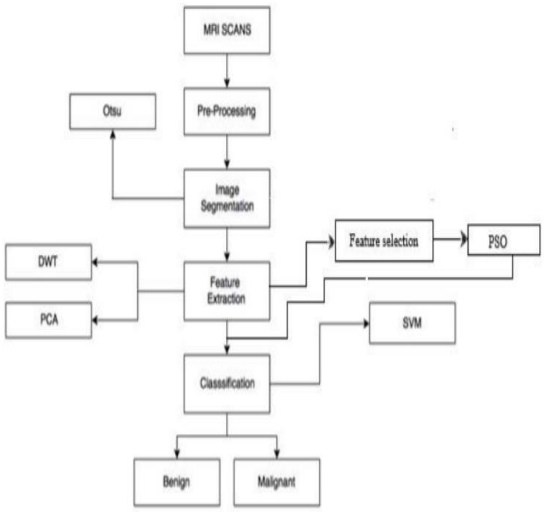


Figure 1: System Architecture

## III. SYSTEM ARCHITECTURE

Magnetic resonance imaging (MRI) is a imaging technique generally used for brain to provide relevant information, based on which physician or CAD can diagnose whether patient have any tumor or not, if there is tumor detected then they can further differentiate between its types so that proper treatment can be given to that patient. Brain tumor mainly in two categories benign and malignant. Benign tumor are non-cancerous while malignant tumor are more prone to be cancerous which may be developed because of cancer at any part of the body not just only brain.

Medical image processing has provided lots of techniques which deals to automate this task in less time and with more accuracy. Feature extraction and selection, Image segmentation and Image classifications are the very important steps in medical image processing. Feature selection is even more important than feature extraction because optimal feature subset is deemed necessary to improve the image classifier performance and decrease the computation time. In traditional approaches we have Principal Component Analysis (PCA) and Linear discriminant Analysis (LDA) for linear, and Kernel. PCA has been used for nonlinear features. In compare to these type of method Evolutionary computation (EC) techniques gives better results. Out of many EC algorithms Particle swarm Optimization is one of the best techniques which can do Feature selection task efficiently with less computation time.

## IV. DATASET COLLECTION

The data collection will contain the predicted data and the actual perceived data, as a percept sequence. The database consists of MRI brain scans taken as input data, that are loaded to the system for training and classification, which are updated itself every time the patient database increases. Only the physician and neurosurgeons will have access rights to modify this database so that sensitive information does not leak. To avoid any latency in operation or lag in performance of the application system, it is necessary to ensure processor is not slow, or that an outdated.

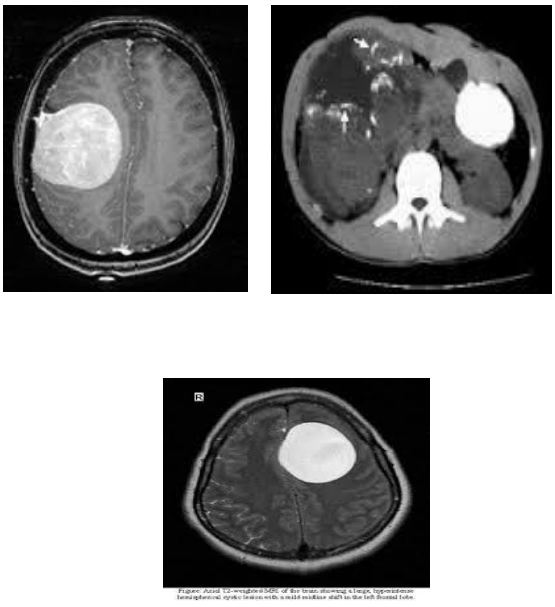

Figure 2: Tumor Images

## V. METHODOLOGY

**1. MRI SCANS:** MRI scans of brain are taken as the input for all training and testing purposes. The images are Grayscale and are resized to 200 x200 px for processing. It is assumed that all the images given input to the system are either Benign (Non-cancerous) or Malignant (Cancerous). The images are separated into two labelled folders for training and testing purpose only. These images are further used for preprocessing.

**2. Pre-processing**: After an MRI image is taken as input image it undergoes into various steps before passing it for image segmentation. First, we have to convert that image into grey scale image then we have to remove noise or inhomogeneity from that grey scale image.

**a. Noise Removal:** First step is conversion of input image in to gray scale image. Gray scale image consists of different levels of gray and it gives 256 possible levels from black to white. Then the unwanted noise can be removed from the gray scale image using median filtering.

**b. Skull Removal:** Skull removal is an important step to detect the brain abnormality. It will increase the processing speed. Noise free brain MRI is taken as the input to skull removal process by morphological operation

**3. Segmentation:** This is the most crucial step in identification of Tumor. This step extracts the tumor from the MRI scans which is then sent for extracting features. Otsu segmentation algorithms that have been implemented.

Otsu's method, named after Nobuyuki Otsu, is used to automatically Perform clustering-based image thresholding. Otsu's thresholding is a non - linear operation that converts a grayscale image into a binary image where the two levels are assigned to pixel those that are below or above the specified threshold value.

**4. Feature extraction:** is the method of dimensionality reduction. It transforms the input data in to set of features which are used for classification purpose. Here we can extract the features using GLCM. It works based on the presence of gray levels present in the image. Entries in GLCM are done by finding relationship between two pixels. GLCM is an 8×8 square matrix and the number of rows and columns equal to the number of pixels in the input image.

**Discrete Wavelet Transform (DWT) and Principal Component Analysis (PCA) with Gray-Level Co-occurrence Matrix Algorithm (GLCM)**

**a.** The **Discrete Wavelet Transform (DWT)** is a powerful implementation of the WT using the dyadic scales and positions. The fundamentals of DWT are introduced as follows. Suppose x(t) is a square-integrable function, then the continuous WT of x(t) relative to a given wavelet ψ(t) is defined as

$$W_\psi(a,b) = \int_{-\infty}^{\infty} x(t)\psi_{a,b}(t)dt$$

$$\psi_{a,b}(t) = \frac{1}{\sqrt{a}}\psi\left(\frac{t-a}{b}\right)$$

**b. GLCM:** is a widely used method for medical image analysis, classification. This method gives us information about relative position of two pixels with respect to each other. The GLCM is then created by counting the number of occurrences of pixel pairs at a certain distance. To compute the GLCM matrix for an image f (i, j), a

distance vector d= (x, y) is defined. The (i,j)th element of the GLCM matrix P is defined as the probability that grey levels i and j occur at distance d and angle θ, then extracting texture features from GLCM matrix P. Four angles (0,45,90,135) and four distances (1,2,3,4) can be used to calculate the co-occurrence matrix.

The extracted features are:

**1. Correlation:** It measures the linear dependency of grey levels of neighbouring pixels.

**2. Contrast:** Also called the sum of Square Variance. It defers the calculation of the intensity contrast linking pixel and its neighbour over the whole image.

**3. Energy:** It makes use for the texture that calculates orders in an image. It gives the sum of square elements in GLCM.

**4. Homogeneity:** It passes the value that calculates the tightness of distribution of the elements in the GLCM to the GLCM diagonal.

**5. Mean:** Defined as the mean of the pixel values of the input image.

**6. Standard Deviation:** It is defined as the dispersion of the pixel in consideration from the mean of the pixels of the input image.

**7. Entropy:** It shows the amount of information of the image that is needed for the image compression. Entropy measures the loss of information or message in a transmitted signal and also measures the image information.

**8. Root Mean Square (RMS):** RMS is calculated on a set of pixels by taking the square of each pixel, calculating the sum of those squares, and taking the square root. The result is scaled by the number of pixels. RMS gives an accurate measurement of the amount of noise present.

**9. Variance:** It is the expectation of the squared deviation of a pixel from its mean.

**10. Inverse Difference Movement (IDM):** Inverse Difference Moment (IDM) is the local homogeneity. It is high when local grey level is uniform and inverse GLCM is high.

**5. FEATURE SELECTION:** Feature selection or reduction is technique in which we select certain feature out of many features which has been extracted based on the property that only these feature leads the effects on the classification process.

**a. Particle Swarm Optimization (PSO):** was originally applied to solve problems in continuous-numbers search space. However, feature selection, as in many other optimization problems, occur in discrete search spaces. Kennedy and Eberhart developed a binary version of the particle swarm algorithm (BPSO) to tackle optimization problems in discrete domains. In BPSO, the velocity is still updated in the same fashion as in the standard PSO. However, variables xid, pid, and pgd can only have the values 0 or 1.

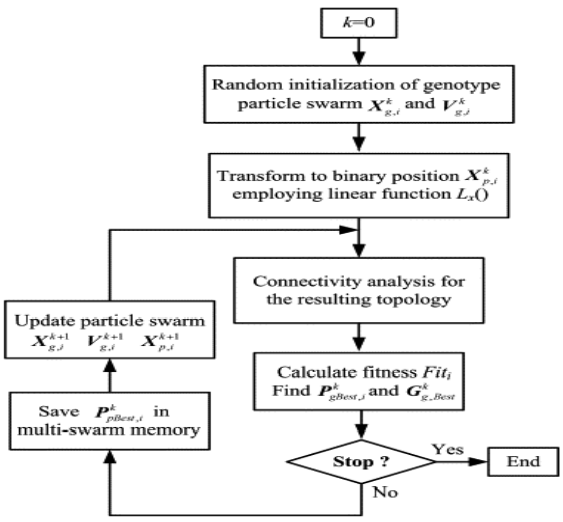


Figure 3: Binary PSO

## 6. CLASSIFICATION

**a. Support Vector Machine (SVM):** Support vector machines (SVMs) are a type of supervised learning models along with associated learning algorithms that analyse data and recognize various patterns, used for classification analysis. The basic SVM takes a set of input data and predicts, for each given input, which of two possible classes, malignant and benign forms the output, making it a non-probabilistic binary linear classifier. Now that there are set of training examples at hand, each marked as belonging to one of two categories, an SVM training algorithm constructs a model that assigns new examples into one category or the other. An SVM model is a representation of the examples as points in space, mapped so that the examples of the separate categories are divided by a clear gap that is as wide as possible. Newer examples are then plotted into it and then predicted to belong to a category based on which side of the gap they fall on.

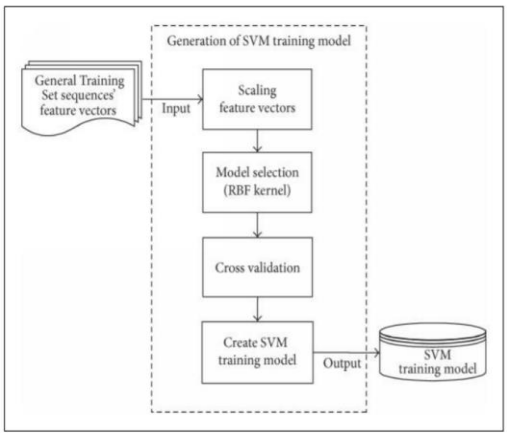


Figure 4:Training SVM Model

**7. TUMOR TYPE:** We have two different type of tumor class benign and malignant in the given dataset. After training SVM classifier with feature selection done by PSO we compare the accuracy of classifier with the case where all the features have been used as Specification and specificity are shown. Total 6 Features out of 14 which are selected by PSO shows the both types of classified tumor benign and malignant tumor.

## VI. EXPERIMENTS AND RESULTS

This project mainly has three modules, Detection of MRI images, Segmentation of the detected tumor region and Detection of tumor type. For our experiments we have considered 30 MRI images i.e. 15 MRI images for Benign tumor and 15 MRI images for Malignant tumor. We have considered 2 separate sets of datasets as training dataset for experiment on detection. The first set contains 20 images of training set and the second set contains 10 images of test set.

We find that accuracy of classification with hybrid model of PSO-SVM is 93.75% which is better than simple SVM classifier which have accuracy of classification 85.7123%.

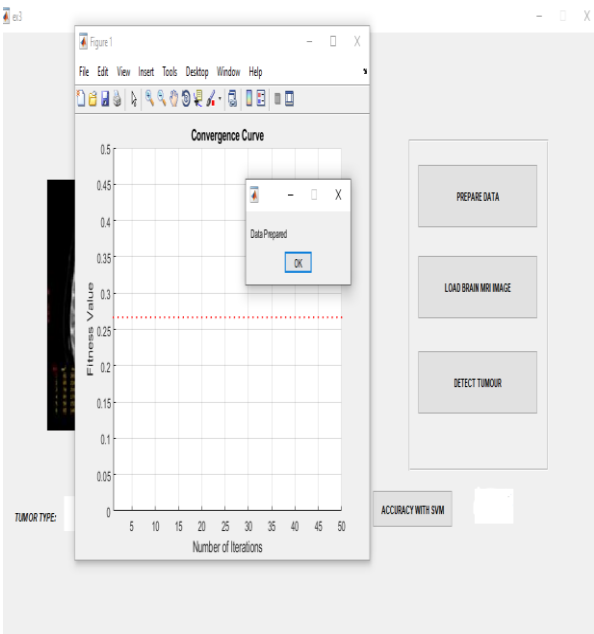


Figure 4: Data Preparation

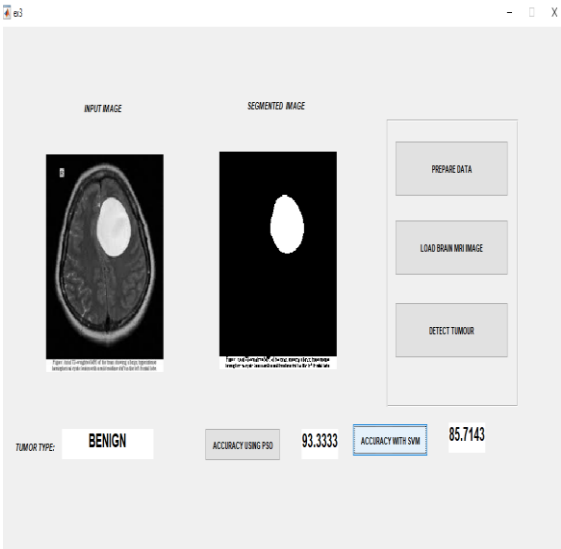


Figure 5: Classification tumor (Benign)

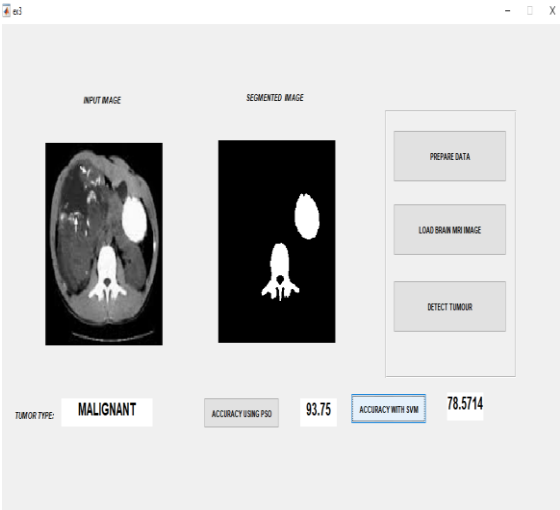


Figure 6: Classification tumor (Malignant)

## VII. CONCLUSIONS

Abnormal growth of tissue in the brain which affect the normal functioning of the brain is considered a brain tumor. The main goal of medical image processing is to identify accurate and meaningful information using algorithms with minimum error possible. Brain tumor detection and classification through MRI images can be categorized into four different sections: pre-processing, image segmentation, feature extraction and image classification. Various segmentation methodologies are explored in the project. It can be concluded that the algorithms and the parameters used in the proposed system are all meant to increase the efficiency of the system by achieving better results.

The boundary approach and the edge-based approach for segmentation are very common but the region growing approach gives better results. It is found that the particle swarm optimization algorithm gives the most accurately segmented tumor. Features extracted by using GLCM method help to increase efficiency as minute details of tumor by using various features can be extracted. Of the various classification methods studied, it was experimentally found that the SVM give the best classification accuracy. Accuracy and reliability are of utmost importance in tumor diagnosis, as a patient's life depends on the results predicted by the system. Thus, the proposed methodology helps in increasing the accuracy and obtaining the desired result.

## VIII. FUTURE ENHNACEMENTS

Encouraged by these results, future work will involve the improvement of classification result and overall accuracy. The number of output classes can also be increased if more data is available. With a more extensive and diverse dataset, the overall classification accuracy can be dramatically increased.

Another approach to improve the result would be to increase the number of hidden layers of the neural network. By increasing the number of hidden layers, the weights will be better adjusted and thus increase the classification. One can also do fine tuning and transfer learning approaches to better tune the model on the basis of already trained models.

## IX. ACKNOWLEDGEMENT

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